

a) GitLab

GitLab ek tool/platform hai jahan developers apna code store aur manage karte hain.

Purpose:

- Code ko store karna.
- Team ke members ke saath code share karna.
- Code changes ko track karna.
- CI/CD ke through code ko automatically test aur deploy karna.

For example:

GitLab ko aap **code ki online storage + management place** samajh sakte ho.

Note: GitLab = Code ko store, manage aur deploy karne ke liye.

b) Kubernetes

Kubernetes ek tool hai jo containers ko manage karta hai.

Purpose:

- Containers ko run karna.
- Containers ko automatically start/stop karna.
- Agar koi container fail ho jaye to usko restart karna.
- Zarurat ke according containers ki quantity badhana ya kam karna.

Note: Agar ek application ke 10 containers chal rahe hain aur ek container fail ho gaya, to Kubernetes usko **automatically restart** kar sakta hai.

Kubernetes = Containers ko manage karne ke liye.

c) ELK Stack

ELK Stack ek logging aur monitoring toolset hai jo system/application ke logs ko collect, store aur easily search karne mein help karta hai.

ELK ka full form:

- **E = Elasticsearch** → Logs ko store aur search karta hai.
- **L = Logstash** → Logs ko collect aur process karta hai.
- **K = Kibana** → Logs ko dashboard/graphs mein dikhata hai.

Purpose:

- Application ke logs collect karna.
- Errors aur problems find karna.
- Logs ko search karna.
- Dashboard ke through information ko easily dekhna.

For example:

Agar application mein error aa gaya, to DevOps Engineer **Kibana** mein logs dekhkar pata laga sakta hai ki problem kya hui.

Note: **ELK = Logs ko collect, search aur visualize karne ke liye.**

d) Puppet

Puppet ek automation tool hai jo servers aur systems ko automatically configure aur manage karta hai.

Purpose:

- Servers ko automatically configure karna.
- Software install/update karna.
- Same configuration multiple servers par apply karna.
- Manual work ko kam karna.

For example:

Agar company ke paas **100 servers** hain aur sabhi mein same software install karna hai, to ek-ek server manually configure karne ke bajay Puppet se ye kaam **automatically** kiya ja sakta hai.

Note: Puppet = Servers ko automatically configure aur manage karne ke liye.